Jayson Boubin

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RESEARCH STATEMENT

I design, implement, and deploy autonomous systems using edge computing to address real societal problems with advanced AI-driven solutions. My intellectual contributions use these systems to create and validate data-driven modeling techniques across disciplines. Over my career, I have used my techniques to build and improve unmanned aerial systems, radar systems, wireless sensor networks, reinforcement learning, deep learning, software bug identification, and edge computing.

EDUCATION

PhD in Computer Science 2017-Present
The Ohio State University

M.S. In Computer Science 2020
The Ohio State University

B.S. in Computer Science 2013-2017
Miami University

WORK EXPERIENCE

PhD Student and Research Fellow 2017-Present
ReRoutLab, Ohio State

At OSU, I developed an open-source platform for autonomous unmanned aerial systems powered by reinforcement learning. I used this platform to study the architectural impacts and resource management implications of single-agent and swarm autonomy. I also built interdisciplinary autonomous UAV applications for agriculture, photography, and outreach.

Research Intern 2019
Air Force Research Lab, AutoWav Project

At AFRL, I built a novel radar waveform design technique using deep learning and neuromorphic hardware acceleration. My approach performed 8X faster than competing neural approaches, with a 40X decreased footprint.

Research Fellow 2014-2017
Air Force Institute of Technology

As a research fellow at AFIT, I performed research in unmanned aerial systems, cybersecurity, applied neuroscience, and autonomy.

Undergraduate Research Assistant 2013-2014
PC2 Lab

I performed research under Dr. Dhananjai Rao focusing on parallel agent based simulation optimization on compute clusters.

AWARDS AND HONORS

NSF Graduate Research Fellowship (2019)

Ohio State College of Engineering Graduate Fellowship (2017)

Best Student Paper, Human Performance Modeling Track,
HFES (2017) (C2)

Best Poster Honorable Mention,
Ohio State CSE Graduate Research Poster Expo (2018) (P6)

KAOC Integrated Product Team Award
AutoWav Project (2019) (P7, C5, C6)

NSF Travel Grants
IEEE ICAC (2017, 2019), IEEE ACSOS (2020), IEEE/ACM SEC (2019)

JOURNAL PAPERS

(J1) Ming-Der Yang, **Jayson Boubin (Corresponding)**, Hui-Ping Tsai, Hsin-Huang Tseng, Yu-Chun Hsu, Christopher Stewart
Adaptive Autonomous UAV Scouting for Rice Lodging Assessment Using Edge Computing with Deep Learning EDANet
Computers and Electronics in Agriculture 2020
(Ranked #3 Journal in Agriculture)

(J2) Zichen Zhang, **Jayson Boubin**, Christopher Stewart, Sami Khanal
Whole-Field Reinforcement Learning: A Fully Autonomous Aerial Scouting Method for Precision Agriculture
Sensors 2020

CONFERENCE PAPERS

(C8) Maxwell Taylor, **Jayson Boubin**, Christopher Stewart, Feng Qin, Haicheng Chen
A Study on Software Bugs in Unmanned Aerial Systems
International Conference on Unmanned Aircraft Systems (ICUAS) 2021
(Ranked #10 Outlet for Aerospace)

(C7) Alwyn Burger, Patrick Urban, **Jayson Boubin**, Gregor Schiele
An Architecture for Solving the Eigenvalue Problem on Embedded FPGAs
International Conference on Architecture of Computing Systems (ARCS) 2020

(C6) Patrick Farr, Aaron Jones, Trevor Bihl, **Jayson Boubin**, Ashley DeMange
Waveform Design Implemented on Neurmorphic Hardware
IEEE International Radar Conference 2020

(C5) **Jayson Boubin**, Aaron Jones, Trevor Bihl
NeuroWav: Toward Real-Time Waveform Design for VANETs using Neural Networks
IEEE Vehicular Networking Conference (VNC) 2019

(C4) **Jayson Boubin**, Naveen T.R Babu, Christopher Stewart, John Chumley, Shiqi Zhang
Managing Edge Resources for Fully Autonomous Aerial Systems
IEEE/ACM Symposium on Edge Computing (SEC) 2019
(Flagship Edge Computing Conference)

(C3) **Jayson Boubin**, John Chumley, Christopher Stewart, Sami Khanal
Autonomic Computing Challenges in Fully Autonomous Precision Agriculture
IEEE International Conference on Autonomic Computing (ICAC) 2019

(C2) **Jayson Boubin**, Christina Rusnock, Jason Bindewald
Quantifying Compliance and Reliance Trust Behaviors to Influence Trust in Human-Automation Teams
Human Factors and Ergonomics Society Annual Meeting (HFES) 2017

(C1) Christina Rusnock, **Jayson Boubin**, Joseph Giametta, Tyler Goodman, Anthony Hillesheim, Sungbin Kim, David Meyer, Michael Watson
The Role of Simulation in Designing Human-Automation Systems
Human Computer Interaction International 2016

PREPRINTS

(PR1) **Jayson Boubin**, Codi Burley, Peida Han, Bowen Li, Barry Porter, Christopher Stewart
Programming and Deployment of Autonomous Swarms using Multi-Agent Reinforcement Learning
ArXiv 2021

PUBLISHED ABSTRACTS

(A2) Bowen Li, Nat Shineman, **Jayson Boubin**, Christopher Stewart
Comparison of Object Detectors for Fully Autonomous Aerial Systems Performance
ACM/SPEC International Conference on Performance Engineering 2021

(A1) **Jayson Boubin**, Shiqi Zhang, Venkata Mandadapu, Christopher Stewart
Poster Abstract: Characterizing Computational Workloads in UAV Applications
ACM/IEE International Conference on Internet of Things Design and Implementation (IoTDI) 2018

POSTERS, PRESENTATIONS, AND DEMOS

(P9) **Jayson Boubin**, Christopher Stewart
Design and Implementation of Fully Autonomous Aerial Swarms
Talk: IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS) 2020 Doctoral Symposium

(P8) **Jayson Boubin**, Christopher Stewart
SoftwarePilot: Fully Autonomous Aerial Systems made Easier
Half-day Tutorial: IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS) 2020

(P7) **Jayson Boubin**, Zichen Zhang, Shiqi Zhang, Christopher Stewart
SoftwarePilot: An Open Source Middleware for Fully Autonomous Aerial Systems

Poster: ACM Student Research Competition, International Symposium on Microarchitecture (MICRO) 2019

(P6) Aaron M. Jones, Trevor Bihl, Ashley DeMange, Peter John-Baptiste, **Jayson Boubin**, Patrick Farr
AutoWav Project: AI/ML for Cognitive EW
Poster: Association of Old Crows Kittyhawk Week Conference

(P5) **Jayson Boubin**, Christina Rusnock
Quantifying and Evaluating Trust in Automated Systems
Talk: Industrial and Systems Engineers Research Conference (ISERC) 2016

(P4) **Jayson Boubin**, Christina Rusnock, Michael Miller
Eliciting an Algorithm to Replicate Human Trust in Automation in the Domain of Reliance
Poster: DESS 2015, Soche Poster expo 2015

(P3) **Jayson Boubin**, Christina Rusnock, Michael Miller
Simulating Compliance and Reliance
Talk: Cincinnati-Dayton INFORMS 2015

(P2) **Jayson Boubin**, Christina Rusnock
Modeling Cognitive Workload and Fatigue for Defensive Cybersecurity Operators
Poster: AFIT Summer Intern Poster Session 2014

(P1) **Jayson Boubin**, Paul Bondurant, D.J. Rao
Dynamic Process Migration in Agent Based Simulation
Poster: Miami University Undergraduate Research Forum 2014

PROFESSIONAL SERVICE

Committee Member:

- Social Media Chair, IoTDI 2022
- Poster and Demo Program Committee member, IoTDI 2020
- Technical Program Committee Member, IEEE DASC 2021

Reviewer:

- Journal of Parallel and Distributed Computing (2021)
- IEEE International Conference on Communications (2021)

Other:

- OHI/O HackAI Hackathon Judge (2019)
- ACM-W Buck-I-Code Mentor (2018)

TEACHING

Graduate Teaching Assistant 2019
The Ohio State University
As and GTA, I was the instructor of record for CSE 2431: Systems II: Introduction to Operating Systems. I taught students about fundamental operating systems concepts including processes, threads, synchronization, memory management, I/O, etc, with a focus on Linux.

I was responsible for lecturing, developing and managing course materials, holding office hours, and submitting and approving grades.

Mean Student Evaluation Score: **4.6/5**

Undergraduate Teaching Assistant 2017

Miami University

As an Undergraduate TA at Miami University, I graded assignments and held weekly help sessions for Dr. Jianhui Yue's Systems 2: Operating Systems course.

ADVISING AND MENTORSHIP

Mentored Graduate Students

Anthony Baietto (2019-Present)

Undergraduate Research Assistants

Jack Dubbs (2019-2020), KeyW Corp
Pieda Han (2019-2020), USC
Yujie Zhao (2019-2020)
Sadaqat Ali (2019-2020), Nationwide
Bowen Li (2019-2021), USC
Nat Shineman (2019-2021)
Chengyuan Zhou (2019-2021), USC
Shiqi Zhang (2017-2019), Morgan Stanley

SOFTWARE

SoftwarePilot

I am the lead developer of SoftwarePilot, a middleware that supports fully autonomous aerial systems. SoftwarePilot has been cloned hundreds of times, and is used by myself, researchers in the ReRoutLab, and a group of scientists and hobbyists around the world to develop state of the art autonomous systems using cutting edge AI libraries. SoftwarePilot is open source and available on Github: *github.com/boubinjg/SoftwarePilot*

SKILLS

Programming (Skilled)	Java, Python, C++
Programming (Comfortable)	C, C#, Matlab, Bash
Operating Systems	Linux
HPC Software	MPI, OpenMP, CUDA
AI Software	Tensorflow, OpenCV, DLIB
UAS Software	DJI SDK, ArduPilot, MAVLink, DroneKit
Version Control	Git, SVN
Editing	Vim, LaTeX
Systems Software	Docker, Kubernetes, Hadoop

REFERENCES

Available upon request